

Principal Component Analysis

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Dimensionality Reduction

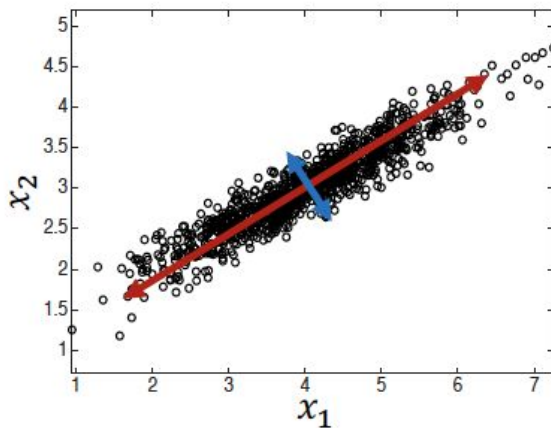
- Curse of Dimensionality
- Finding a lower number of variables that can describe the variability in a dataset

Dimensionality Reduction vs Subset Variable Selection

- Not The same
- Variable selection is with respect to a target (classification) variable
- In dimensionality reduction, there is no target variable

PCA

- Principal Component Analysis (PCA) finds linear combinations of orthogonal variables that explain the maximal amount of variance in the data.
- PCA finds an alternative vector basis for the data, where
 - the 1st dimension explains the maximal amount of variance in the data
 - the 2nd dimension explains the maximal amount of variance left to explain (orthogonal to the 1st dimension), and so on

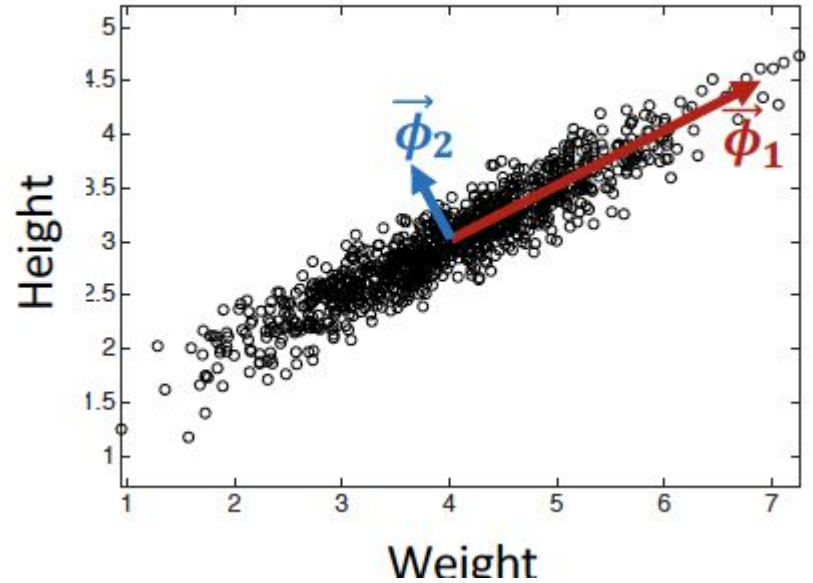


PCA

$$\mathbb{R}^{n \times p} \ni \vec{x} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,p} \end{bmatrix}$$

$$\mathbb{R}^n \ni \vec{z}_1 = \phi_{1,1}\vec{x}_1 + \phi_{2,1}\vec{x}_2 + \cdots + \phi_{p,1}\vec{x}_p = \vec{x} \cdot \vec{\phi}_1.$$

$$\vec{\phi}_1 = (\phi_{1,1}, \phi_{2,1}, \dots, \phi_{p,1})^T \in \mathbb{R}^p$$



First Principal Component

- The first principal component is the normalized linear combination of the features resulting in the maximal variance
 - If it is not normalized then the maximization does not make sense (can be ∞)

$$\sum_{j=1}^p \phi_{j,1}^2 = 1 \quad \text{or} \quad \|\vec{\phi}_1\| = 1.$$

First Principal Component

- We subtract the mean vector to make it zero-centered

$$\mathbb{R}^n \ni \vec{z}_1 = \phi_{1,1}\vec{x}_1 + \phi_{2,1}\vec{x}_2 + \dots + \phi_{p,1}\vec{x}_p = \vec{x} \cdot \vec{\phi}_1.$$

$$\vec{\phi}_1 = \operatorname{argmax}_{\|\vec{\phi}_1\|=1} \|\vec{z}_1\|^2 \quad (\text{subject to } \|\vec{\phi}_1\| = 1),$$

which is

$$\vec{\phi}_1 = \operatorname{argmax}_{\|\vec{\phi}_1\|=1} \|\vec{x} \cdot \vec{\phi}_1\|^2$$

$$= \operatorname{argmax}_{\|\vec{\phi}_1\|=1} \left\{ \vec{\phi}_1^T \vec{x}^T \vec{x} \vec{\phi}_1 \right\}_{\|\vec{\phi}_1\|=1} = \operatorname{argmax} \left\{ \frac{\vec{\phi}_1^T \vec{x}^T \vec{x} \vec{\phi}_1}{\vec{\phi}_1^T \vec{\phi}_1} \right\}.$$

First Principal Component

$$\text{cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{x})(Y_i - \bar{y})$$

$$\vec{C} = \vec{x}^T \vec{x}$$

$$\frac{\vec{\phi}_1^T \vec{x}^T \vec{x} \vec{\phi}_1}{\vec{\phi}_1^T \vec{\phi}_1} = \frac{\vec{\phi}_1^T \vec{C} \vec{\phi}_1}{\vec{\phi}_1^T \vec{\phi}_1}$$

First Principal Component

- $\lambda_{\max}$ and $\vec{v}_{\max}$ are the maximum eigenvalue of C and its corresponding eigenvector

$$\frac{\vec{\phi}_1^T \vec{C} \vec{\phi}_1}{\vec{\phi}_1^T \vec{\phi}_1} \leq \lambda_{\max}.$$

$$\frac{\vec{v}_{\max}^T \vec{C} \vec{v}_{\max}}{\vec{v}_{\max}^T \vec{v}_{\max}} = \lambda_{\max}.$$

Second , Third and other Principal Components

- Calculate the Principal component for the remaining variance

$$\vec{x}_{(m)} = \vec{x} - \sum_{i=1}^{m-1} \vec{x} \vec{\phi}_i \vec{\phi}_i^T ,$$

$$\vec{\phi}_m = \operatorname{argmax}_{\|\vec{\phi}_m\|=1} \|\vec{x}_{(m)} \cdot \vec{\phi}_m\|^2 = \operatorname{argmax}_{\|\vec{\phi}_m\|=1} \left\{ \frac{\vec{\phi}_m^T \vec{x}_{(m)} \vec{x}_{(m)}^T \vec{\phi}_m}{\vec{\phi}_m^T \vec{\phi}_m} \right\} .$$

PCA

$$\vec{z} = \vec{x}\vec{\phi},$$

$$\mathbb{R}^{p \times p} \ni \vec{\phi} = [\vec{\phi}_1, \vec{\phi}_2, \dots, \vec{\phi}_p],$$

$\vec{\phi}_k$ is the k 'th eigenvector of $\vec{x}^T \vec{x} = \vec{C}$ (the covariance matrix of $\vec{x}$).

PCA

- Each principal component is a combination of all other variables
- For example, if we have two variables like age and income in a dataset, then the first principal component might be a combination like this:

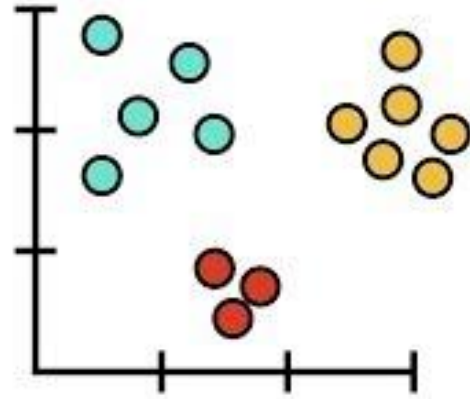
$$0.4\text{age} + 0.6\text{income}$$

- How do you interpret this new variable? Is it age? Is it income? Or is it something between age and income?

How to perform PCA: Steps

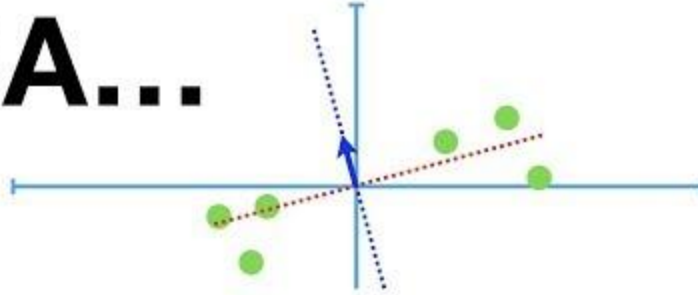
- Take the whole dataset consisting of $d+1$ dimensions and ignore the labels such that our new dataset becomes d dimensional.
- Compute the mean vector and subtract
- Compute the covariance matrix
- Compute eigenvectors and the corresponding eigenvalues of the Covariance Matrix.
- Sort the eigenvectors by decreasing eigenvalues and choose k eigenvectors with the largest eigenvalues to form a $d \times k$ dimensional matrix W .
- Use this $d \times k$ eigenvector matrix to transform the samples onto the new subspace.

PCA Main Ideas...



..in only 5 min!!!

PCA...



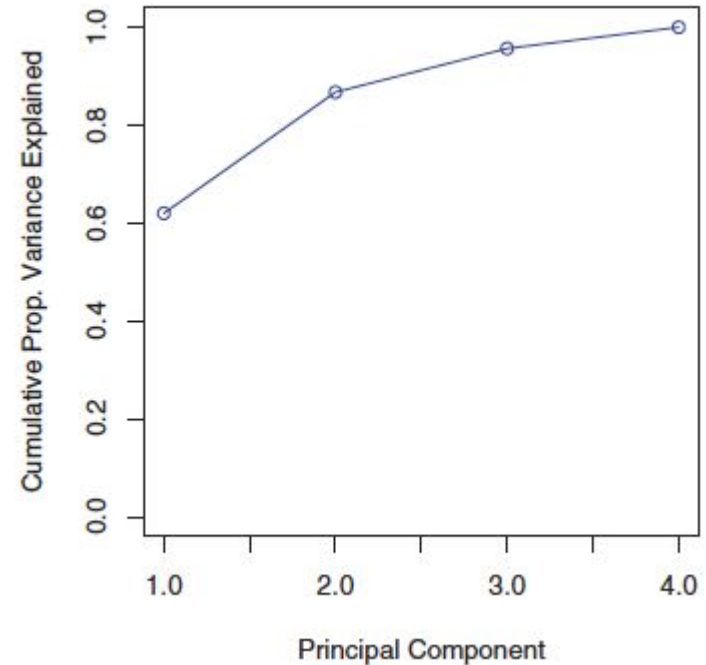
Step-by-Step!!!

<https://www.youtube.com/watch?v=FgakZw6K1QQ>

How many Dimensions are good enough?

Elbow method

$$\frac{\sum_{i=1}^q \lambda_i}{\sum_{i=1}^p \lambda_i}.$$



Advantages of Principal Component Analysis

- Unsupervised preprocessing technique is labels agnostic, hence does not lead to double dipping
- Dimensionality reduction may lead to data that is easier to visualize
- Running PCA does not rest on any special assumptions

Disadvantages of Principal Component Analysis

- **Principal components (linear combinations) can also sometimes be difficult to interpret directly**
- Nonlinear structure can be hard to model with PCA (though nonlinear PCA methods with kernels and other extensions do exist)